

USER INTERFACES LAYER

CLI

cli.py

Web UI

app.py

API

api.py

APPLICATION LAYER

DataProcessor (Core Engine)

- read\_file()
- extract\_discrepancy()

- compare\_files()
- classify\_discrepancies()

Validator

Rule Engine

JobManager

Orchestrator

UIHelpers

UI Components



DATA ACCESS LAYER

FileHandler

File I/O

DatabaseHandler

DB Operations

Visualizer

Charts/Graphs



DATA SOURCES

Excel Files

(.xlsx, .xls)

CSV Files

(.csv)

Text Files

(.txt, .log)

Databases

(PostgreSQL,  
MySQL, Oracle, MSSQL)



OUTPUT LAYER

Discrepancy Results

(DataFrame)

Reports

(CSV/JSON/XLSX)

Visualizations

(Plotly Charts)